

Algorithms

Lecture # 35

Syed Hamed Raza

Minimum Spanning Trees

Minimum Spanning Trees

Free Tree Facts

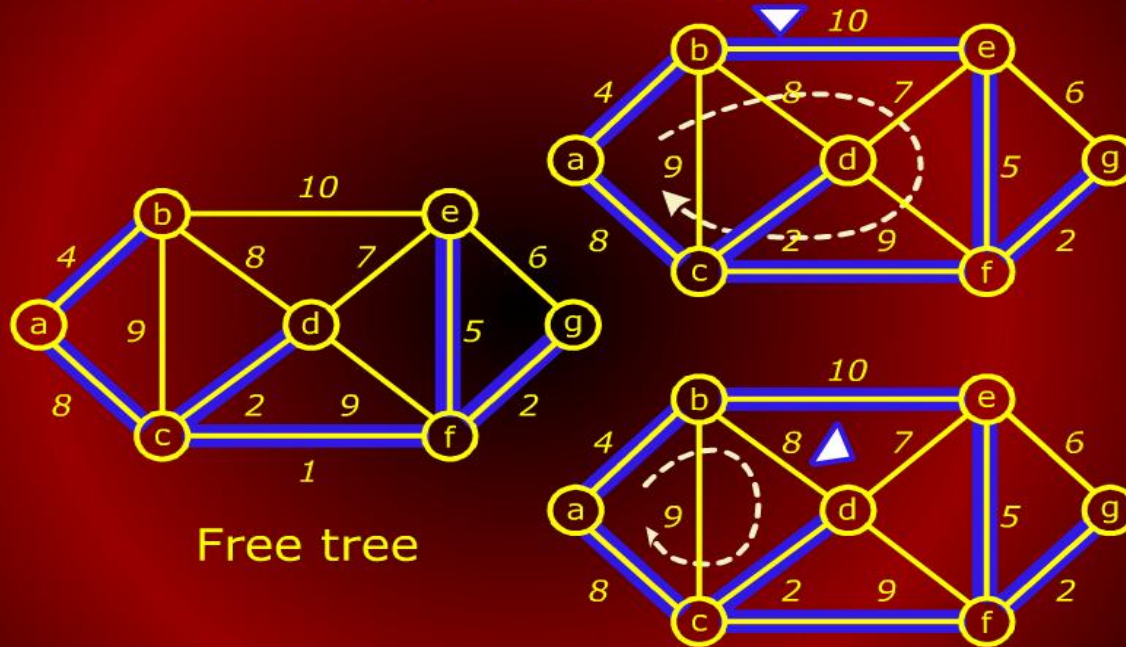
Free Tree Facts

Before presenting the two algorithms, let us review facts about free trees:

- A free tree with n vertices has exactly $n - 1$ edges.
- There exists a unique path between any two vertices of a free tree.
- Adding any edge to a free tree creates a unique cycle. Breaking any edge on this cycle restores the free tree.

Free Tree Facts

Free Tree Facts



Generic Approach

Generic Approach

- Let $G = (V, E)$ be an undirected, connected graph whose edges have numeric weights.
- The intuition behind greedy MST algorithm is simple: we maintain a subset of edges A .
- Initially, A is empty.
- We will add edges one at a time until A equals the MST.

Generic Approach

Generic Approach

- A subset $A \subseteq E$ is *viable* if A is a subset of edges of *some* MST.
- An edge $(u, v) \in E - A$ is safe if $A \cup \{(u, v)\}$ is viable.
- In other words, the choice (u, v) is a safe choice to add so that A can still be extended to form a MST.

Generic Approach

Generic Approach

- Note that if A is viable, it cannot contain a cycle.
- A generic greedy algorithm operates by repeatedly adding any safe edge to the current spanning tree.

Generic Approach

Generic Approach

When is an edge safe?

- Consider the theoretical issues behind determining whether an edge is safe or not.
- Let S be a subset of vertices $S \subseteq V$.
- A cut $(S, V - S)$ is just a partition of vertices into two disjoint subsets.
- An edge (u, v) crosses the cut if one endpoint is in S and the other is in $V - S$.

Generic Approach

Generic Approach

- Given a subset of edges A , a cut respects A if no edge in A crosses the cut.
- It is not hard to see why respecting cuts are important to this problem.
- If we have computed a partial MST and we wish to know which edges can be added that do not induce a cycle in the current MST, any edge that crosses a respecting cut is a possible candidate.

Greedy MST Proof

Greedy MST

- An edge of E is a light edge crossing a cut if among all edges crossing the cut, it has the minimum weight.
- Intuition says that since all the edges that cross a respecting cut do not induce a cycle, then the lightest edge crossing a cut is a natural choice.
- The main theorem which drives both algorithms is the following:

Greedy MST Proof

Greedy MST

MST Lemma:

- Let $G = (V, E)$ be a connected, undirected graph with real-valued weights on the edges.
- Let A be a viable subset of E (i.e., a subset of some MST).
- Let $(S, V - S)$ be any cut that respects A and let (u, v) be a light edge crossing the cut.
- Then the edge (u, v) is safe for A .

Greedy MST Proof

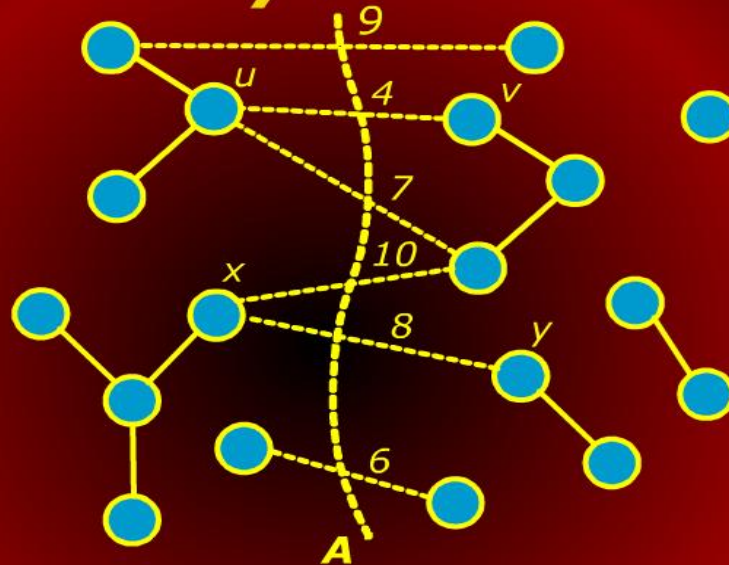
Greedy MST Proof

MST Proof:

- It would simplify the proof if we assume that all edge weights are distinct.
- Let T be any MST for G .

Greedy MST Proof

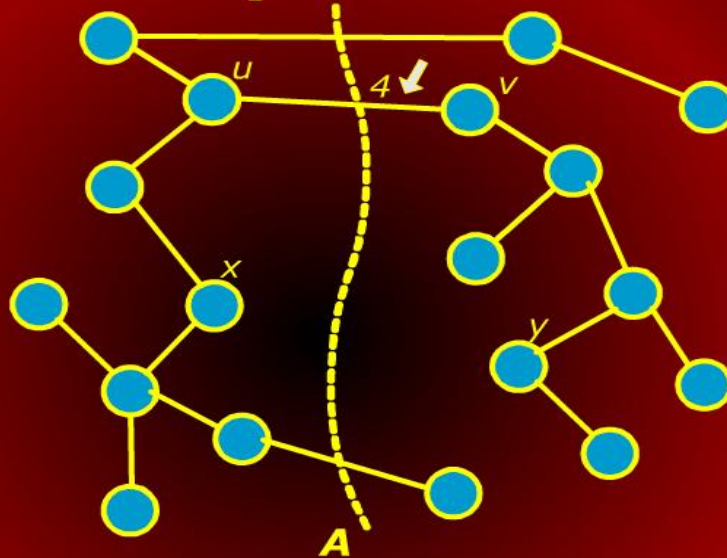
Greedy MST Proof



Let T be any MST for G .

Greedy MST Proof

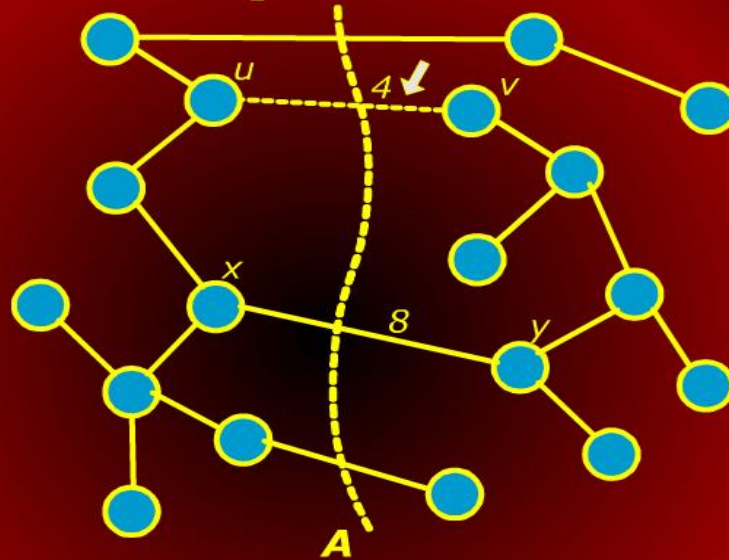
Greedy MST Proof



If T contains (u, v) then we are done.

Greedy MST Proof

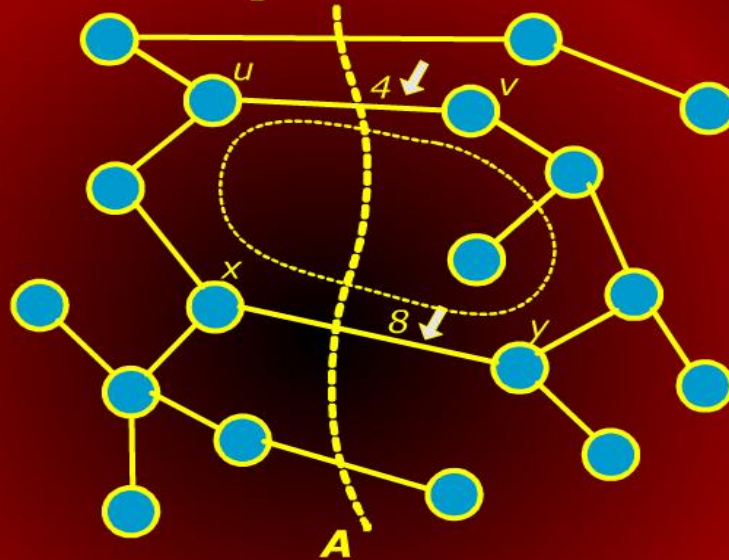
Greedy MST Proof



Suppose no MST contains (u, v) . We will derive a contradiction.

Greedy MST Proof

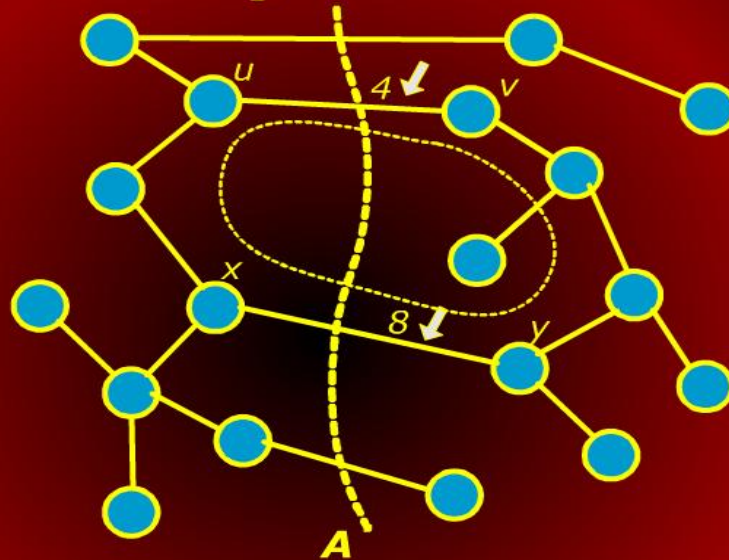
Greedy MST Proof



Add (u, v) to T thus creating a cycle.

Greedy MST Proof

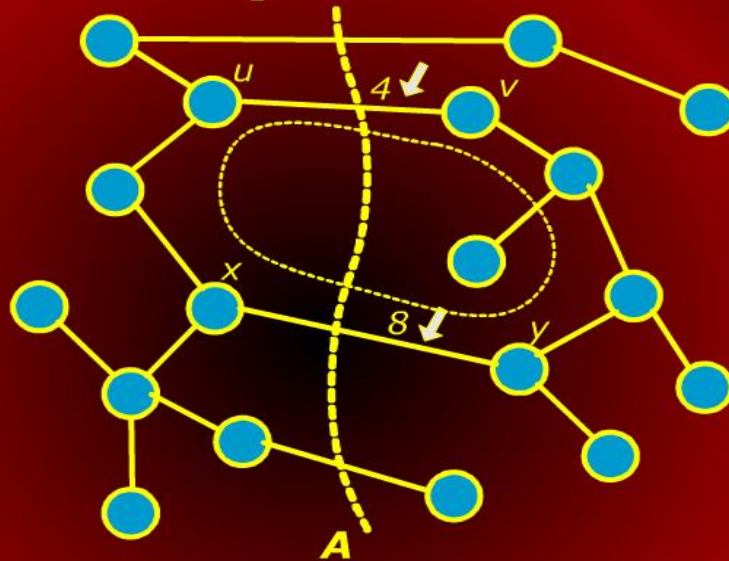
Greedy MST Proof



Since u and v are on opposite sides of the cut, and

Greedy MST Proof

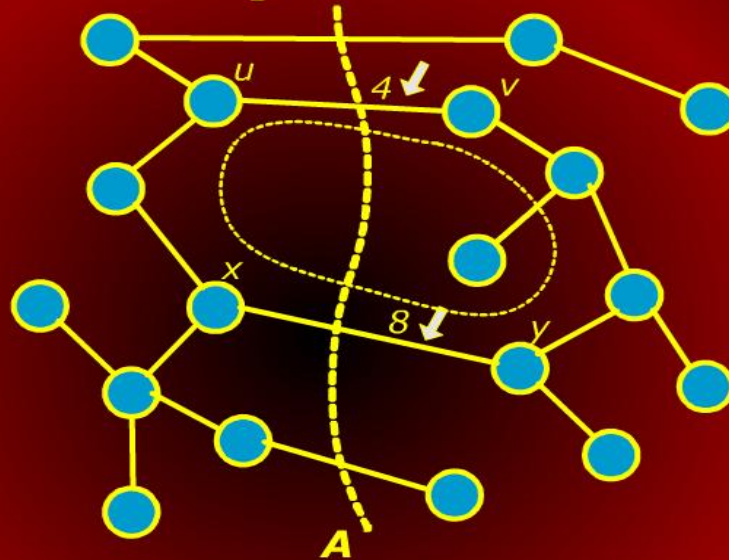
Greedy MST Proof



since any cycle must cross the cut an even number of times,

Greedy MST Proof

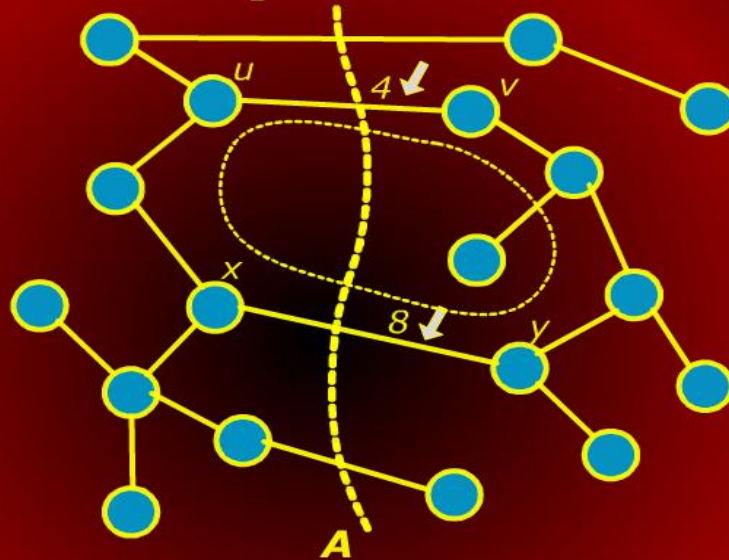
Greedy MST Proof



there must be at least one other edge (x, y) in T that crosses the cut.

Greedy MST Proof

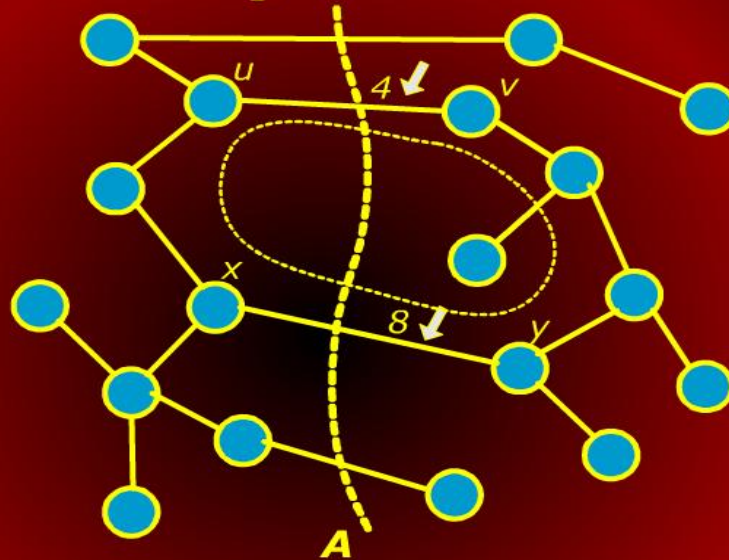
Greedy MST Proof



The edge (x, y) is not in A because the cut respects A .

Greedy MST Proof

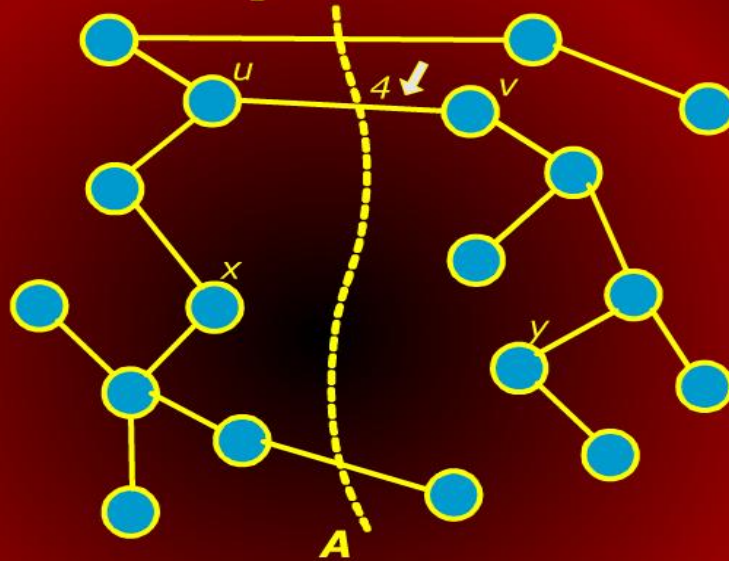
Greedy MST Proof



By removing (x, y) we restore a spanning tree, call it T' .

Greedy MST Proof

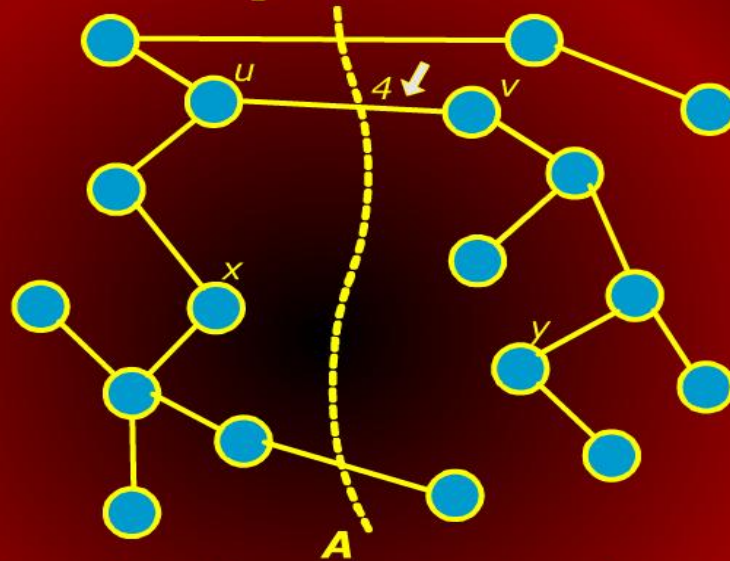
Greedy MST Proof



We have $w(T') = w(T) - w(x, y) + w(u, v)$.

Greedy MST Proof

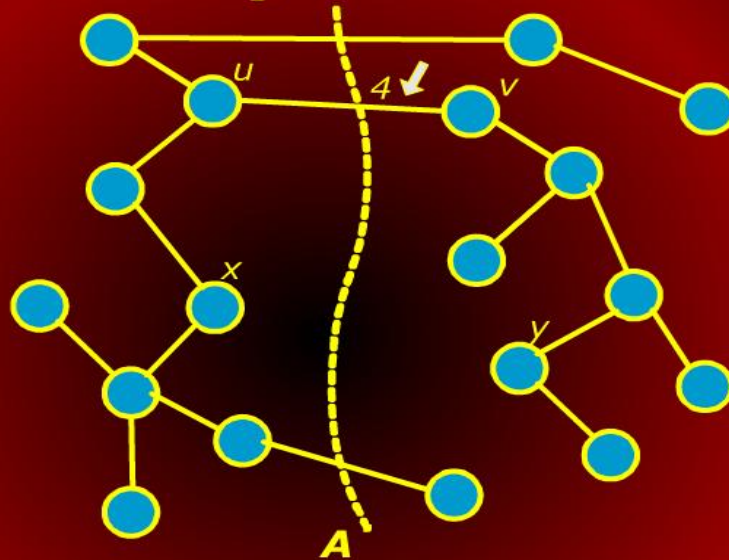
Greedy MST Proof



Since (u, v) is the lightest edge crossing the cut we have $w(u, v) < w(x, y)$.

Greedy MST Proof

Greedy MST Proof



Thus $w(T') < w(T)$ which contradicts the assumption that T was an MST.

Random Access Machine